

KRISTIINANKAUPUNKI FYR, Finland

WMO: 027520

Lat: 62.2036N

Lon: 21.1697E

Elev: 71

StdP: 14.66

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.5	7.8	-1.5	5.1	3.6	3.4	6.6	9.3	43.9	31.3	40.4	30.2	10.4	50	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.6	73.5	65.6	71.0	64.4	68.2	62.5	68.0	72.2	65.5	69.2	63.6	67.2	12.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	96.7	70.7	64.0	89.6	68.1	62.0	83.3	66.0	32.4	72.4	30.5	69.3	29.0	67.2	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
36.7	33.0	29.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.5	25.6	26.1	29.6	36.4	47.0	53.6	61.3	61.0	55.0	44.0	37.3	31.7
	DBStd	14.16	8.28	9.29	5.82	4.24	7.27	5.67	5.56	4.25	4.40	5.33	6.93	8.43
	HDD50	3794	757	669	631	410	149	21	0	1	14	193	382	567
	HDD65	8270	1222	1089	1096	858	560	346	146	138	301	650	832	1032
	CDD50	1054	0	0	0	2	56	130	352	342	164	8	0	0
	CDD65	53	0	0	0	0	1	5	32	15	0	0	0	0
	CDH74	80	0	0	0	0	2	4	46	28	0	0	0	0
	CDH80	7	0	0	0	0	0	0	4	3	0	0	0	0
Wind	WSAvg	15.4	16.2	16.3	13.8	14.1	14.6	14.0	12.8	13.5	15.0	17.3	18.3	19.2
Precipitation	PrecAvg	24.4	1.8	1.3	1.3	1.1	1.6	2.2	2.9	2.9	2.2	2.6	2.3	2.2
	PrecMax	31.3	3.9	2.8	2.2	2.4	3.5	4.5	5.3	6.2	5.5	6.1	5.0	4.2
	PrecMin	18.1	0.3	0.1	0.2	0.1	0.2	0.6	0.4	0.6	0.8	0.4	0.4	0.6
	PrecStd	3.2	0.8	0.7	0.5	0.6	0.8	0.9	1.2	1.3	1.3	1.2	1.0	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.3	39.6	42.8	58.2	72.3	73.9	78.9	78.2	65.6	56.8	50.7	45.7
		MCWB	37.7	36.7	37.5	46.8	56.9	61.3	66.8	66.3	60.1	53.2	48.3	43.5
	2%	DB	38.7	38.2	39.3	46.8	66.5	69.5	74.3	71.9	62.6	54.0	48.5	43.7
		MCWB	36.4	35.8	36.4	41.6	57.1	59.6	66.4	66.0	58.9	51.4	47.2	41.5
	5%	DB	36.9	36.9	37.8	43.1	61.8	65.1	72.0	69.1	61.5	52.5	46.6	41.8
		MCWB	35.2	35.0	35.6	38.8	54.9	58.4	66.0	63.8	58.1	49.9	44.8	39.5
	10%	DB	35.4	35.6	36.5	40.9	57.1	61.3	69.7	66.3	60.3	50.9	45.0	40.3
		MCWB	34.0	34.3	34.6	38.0	50.9	57.1	64.3	62.0	57.1	48.3	43.3	38.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.4	37.7	39.1	47.2	60.7	63.8	70.5	71.2	61.1	54.0	48.6	44.2
		MCDB	40.0	39.1	41.0	56.9	65.6	68.4	74.6	74.9	63.9	55.7	50.0	45.2
	2%	WB	36.9	36.5	37.1	42.3	57.9	61.2	68.9	66.6	60.0	52.2	47.4	41.7
		MCDB	38.4	37.9	38.5	45.7	65.5	66.5	73.2	70.6	62.1	53.8	48.3	43.3
	5%	WB	35.7	35.6	36.0	40.0	54.9	59.4	67.0	64.7	58.9	50.6	45.4	39.8
		MCDB	37.0	36.9	37.5	42.5	61.7	64.5	70.9	68.3	61.1	52.1	46.3	41.4
	10%	WB	34.3	34.6	35.1	38.5	51.4	57.1	64.8	62.9	57.9	48.8	43.5	38.6
		MCDB	35.4	35.8	36.5	40.3	56.1	61.2	68.5	66.1	59.9	50.5	44.6	40.1
Mean Daily Temperature Range	5% DB	MDBR	7.0	7.0	6.6	6.1	7.9	6.9	6.6	5.5	5.3	5.1	5.0	5.4
		MCDBR	6.0	6.4	6.4	10.0	14.2	13.1	10.8	8.0	6.3	4.7	4.8	5.2
		MCWBR	5.4	5.5	5.1	6.1	6.9	7.0	6.1	5.5	4.9	5.0	5.0	5.3
	5% WB	MCDBR	6.0	5.9	6.0	9.4	13.1	11.3	8.5	6.1	5.4	4.8	5.0	5.0
		MCWBR	5.5	5.1	5.3	6.4	6.8	7.2	6.0	5.2	4.6	5.3	5.3	5.3
Clear-Sky Solar Irradiance	taub	0.255	0.304	0.322	0.361	0.359	0.353	0.377	0.364	0.340	0.323	0.273	0.349	
	taud	2.119	2.239	2.320	2.338	2.432	2.485	2.442	2.467	2.515	2.392	2.166	2.538	
	Ebn at Noon	133	206	250	262	273	277	266	259	245	200	123	81	
	Edh at Noon	11	20	26	32	32	30	31	28	22	17	10	7	
All-Sky Solar Radiation	RadAvg	53	228	707	1221	1735	1922	1757	1359	806	319	79	30	
	RadStd	7	43	83	129	156	118	145	115	75	58	15	3	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (43 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	-218	N/A	N/A	N/A

Nomenclature: See separate page